Introduction to the theoretical foundations of linear gap models

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OGResearch

Motivation

The RBC model structure

The RBC model structure

General properties

Consumers

Firms

Ingredients of a general equilibrium model

- We explicitly specify the *economic agents* in the model
- We also specify the *goods* which these agents produce or absorb (consume)
- We make assumptions about the *markets* on which the agents interact
- We assume rules which govern their *economic behavior*
- These rules must respect *physical constraints*, as well as *budget constraints*
- Equilibrium conditions ensure that optimal demand and supply are equal

Structure of the basic RBC model

- Two representative agents: consumer and firm
- Two goods: labor and a consumption/investment good
- Agents are price takers on competitive markets
- Consumers and firms behave optimally
- Spending plus saving of consumer can not exceed their income
- A production function relates output to capital and labor
- Capital depreciates over time, and it is produced from investment
- Consumption plus investment is equal to output

Consumers

- The consumer
  - Absorbs the consumption good, and supplies labor
  - Owns the capital stock, which is rented to the firms
  - Consequently, consumers make the investment decisions, and they face the capital accumulation constraint
  - Adheres to a budget constraint consisting of labor income, consumption and investment cost, and savings
  - Also faces the capital accumulation constraint
  - Is infinitely-lived, and maximizes *expected lifetime utility*

Firms

- The firm
  - Absorbs labor, capital
  - Supplies consumption/investment and capital
  - Maximizes the discounted value of expected profit
  - Is constrained by the production function

Perfect competition

- Perfectly competitive markets mean that
  - Both agents are price takers
  - So that prices are fixed in their optimizing decisions
  - Prices adjust in an *unspecified way* to achieve equilibrium
- In equilibrium, optimal demand and supply of all goods are equal

The RBC model structure

General properties

Consumers

Firms

Properties of consumer behavior

- We want to capture the following properties of consumer behavior
  - Consumers are *forward looking*
    - They care about utility in all future periods, but with less weight given to more distant future
  - They like consumption, and they (possibly) dislike labor
  - The *marginal utility of consumption is positive and declining*
    - An extra unit of consumption is better when consumption is still low
  - The *marginal disutility of labor is positive and increasing*
    - An extra unit of labor is worse when labor is already high

Lifetime utility

- These considerations lead to the following choice of the utility function (assuming 0 stands for the current period)

$$\sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \theta \, \frac{L_t^{1+\eta}}{1+\eta} \right) \tag{1}$$

- C_t is real consumption, L_t is labor in period t
- $0 < \beta < 1$ is the *subjective discount factor*
- $0 < \sigma$ is the *intertemporal elasticity of substitution*
- $0 < \eta$ is the *elasticity of substitution of labor*
- $0 < \theta$ is a scaling factor for labor
  - Depends on the unit of labor (i.e. years, days, hours)

Marginal utilities

- The marginal utility of consumption in period t is $C_t^{-\sigma}$, which is positive and decreasing when $0 < \sigma$
- The marginal disutility of labor in period t is $\theta\, L_t^{\eta}$, which is positive and increasing when $0 < \eta$
- $\beta < 1$ implies that β^t is decreasing

The budget constraint

- Consumers receive wage and spend on consumption and investment
- They may have savings, stored in a financial asset or physical capital
- The interest on previous savings is another source of spending
- In formula

$$P_t\, C_t + P_t\, I_t + B_t = W_t\, L_t + i_{t-1}\, B_{t-1} + P_t^K K_{t-1} \qquad \text{for } 0 \le t \qquad (2)$$

- Here

P_t	is the price of consumption in period t
W_t	is the nominal wage in period t
i_t	is the nominal gross interest rate at the end of period t
P_t^K	is the rental price of capital in period t

The optimization problem of consumers

- The objective function of the consumer is

$$\max_{C_t, L_t, B_t} \sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \theta \, \frac{L_t^{1+\eta}}{1+\eta} \right) \tag{3}$$

- Their constraints are

$$P_t \, C_t + P_t \, I_t + B_t = W_t \, L_t + i_{t-1} \, B_{t-1} + P_t^K K_{t-1} \tag{4}$$

$$K_t = (1-\delta) \, K_{t-1} + I_t \tag{5}$$

- The constraint holds for all $0 \leq t$
- The consumers take B_{-1}, K_{t-1}, P_t, W_t, i_t, P_t^K as given

The Lagrangian

- The Lagrangian of the constrained optimization problem is

$$
\begin{aligned}
\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Bigg[\left(\frac{C_t^{1-\sigma}}{1-\sigma} - \theta \, \frac{L_t^{1+\eta}}{1+\eta} \right) \\
-\Lambda_t \left(P_t \, C_t + P_t \, I_t + B_t - \left(W_t \, L_t + i_{t-1} \, B_{t-1} + P_t^K \, K_{t-1} \right) \right) \\
-Q_t \, \Lambda_t \left(K_t - \left((1-\delta) \, K_{t-1} + I_t \right) \right) \Bigg]
\end{aligned}
\tag{6}
$$

- Note that β^t multiplies Λ_t
- We are free to define the Lagrange multiplier in terms of parameters and fixed variables

Interpretation of Λ_t

- In optimum, the consumer must be indifferent to small re-allocations of consumption and labor
- If consumption is C_t, a small decrease ε in consumption has utility loss $C_t^{-\sigma}\,\varepsilon$
- On the other hand, we have more money to spend, given by $P_t\,\varepsilon$
- In equilibrium, the *subjective value* of the extra wealth must be equal to the utility loss
- Λ_t translates a unit of extra wealth into utility units, so that we get
$$C_t^{-\sigma}\,\varepsilon = \Lambda_t\,P_t\,\varepsilon$$
- We say Λ_t is the *shadow price of wealth*

Optimal decision

- Optimality conditions are

$$\beta^t \left(C_t^{-\sigma} - \Lambda_t\, P_t\right) = 0 \qquad \left(\frac{\partial}{\partial C_t}\right)$$

$$\beta^t \left(-\theta\, L_t^{\eta} + \Lambda_t\, W_t\right) = 0 \qquad \left(\frac{\partial}{\partial L_t}\right)$$

$$-\beta^t\, \Lambda_t + \beta^{t+1}\, \Lambda_{t+1}\, i_t = 0 \qquad \left(\frac{\partial}{\partial B_t}\right)$$

$$\beta^t \left(-\Lambda_t\, P_t + Q_t\, \Lambda_t\right) = 0 \qquad \left(\frac{\partial}{\partial I_t}\right)$$

$$-\beta^t\, Q_t\, \Lambda_t + \beta^{t+1}\, \Lambda_{t+1}\, P_{t+1}^K + \beta^{t+1}\, Q_{t+1}\, \Lambda_{t+1}\, (1-\delta) = 0 \qquad \left(\frac{\partial}{\partial K_t}\right)$$

Rearranging the equations

- Rearrange:

$$C_t^{-\sigma} = \Lambda_t \, P_t \tag{7}$$

$$\theta \, L_t^{\eta} = \Lambda_t \, W_t \tag{8}$$

$$\Lambda_t = \beta \, \Lambda_{t+1} \, i_t \tag{9}$$

$$Q_t = P_t \tag{10}$$

$$\Lambda_t \, Q_t = \beta \, \Lambda_{t+1} \, P_{t+1}^K + \beta \, \Lambda_{t+1} \, Q_{t+1} \, (1-\delta) \tag{11}$$

- Combining the last two equations

$$\Lambda_t \, P_t = \beta \, \Lambda_{t+1} \, P_{t+1}^K + \beta \, \Lambda_{t+1} \, P_{t+1} \, (1-\delta) \tag{12}$$

The RBC model structure

General properties

Consumers

Firms

Profit maximization

- The firm's objective function is

$$\max_{Y_t, K_{t-1}, L_t} \sum_{t=0}^{\infty} D_t \left(P_t\, Y_t - P_t^K\, K_{t-1} - W_t\, L_t\right) \tag{13}$$

- The only constraint is

$$Y_t = K_{t-1}^{\alpha} \left(A_t\, L_t\right)^{1-\alpha} \tag{14}$$

- This is not a dynamic optimization problem, since today's decisions does not influence tomorrow's problem
- The equations will be

$$RP_t^K = \alpha\, \frac{Y_t}{K_{t-1}} \tag{15}$$

$$WR_t = (1-\alpha)\, \frac{Y_t}{L_t} \tag{16}$$

Consumers owning the capital

- The model will be (keeping the "raw" first order conditions including Λ_t)

$$\Lambda_t = \beta\, \Lambda_{t+1}\, R_t \tag{17}$$

$$C_t^{-\sigma} = \Lambda_t \tag{18}$$

$$\theta\, L_t^{\eta} = \Lambda_t\, WR_t \tag{19}$$

$$\Lambda_t = \beta\, \Lambda_{t+1} \left(RP_{t+1}^{K} + (1-\delta) \right) \tag{20}$$

$$RP_t^{K} = \alpha\, \frac{Y_t}{K_{t-1}} \tag{21}$$

$$WR_t = (1-\alpha)\, \frac{Y_t}{L_t} \tag{22}$$

$$Y_t = C_t + I_t \tag{23}$$

$$K_t = (1-\delta)\, K_{t-1} + I_t \tag{24}$$

$$Y_t = K_{t-1}^{\alpha}\, (A_t\, L_t)^{1-\alpha} \tag{25}$$

Price setting

Monetary policy

- Adding monetary policy leads to two new variables: inflation $\mathrm{d}P_t$, and nominal interest rates RN_t
- So we need two more equations
- If we want any connection between the real economy and monetary policy, we need to relate the new variables to existing ones
- That relation is the Fisher equation

$$R_t = \frac{RN_t}{\mathrm{d}P_{t+1}} \tag{M.1}$$

A missing variable

- We can add only one more equation
- Either for the nominal rate, or for inflation
- That means that one will be always determined from the other by the Fisher equation
- *Without changing the existing model, the real interest rate will be determined independently of monetary policy*
- We need a relationship between the real economy and inflation, in other words a Phillips curve
- And the notion of *price pressure*
- This will only be possible if have explicit price setters in the model

Stages of production

- We have a three-stage production process
  - The final good Y_t is aggregated by a competitive firm, from intermediate inputs $Y_{t,i}$
  - The intermediate good $Y_{t,i}$ is distributed by distributors, with some form of market power, and adjustment cost for changing prices
  - The intermediate good $Y_{t,i}$ is produced from capital and labor as before
- The distributors are not "fully rational"
  - They recognize that their price effects demand for their specific good i
  - But they *do not recognize that their decisions also effect aggregate prices and demand*

Aggregator I

- They produce Y_t from $Y_{t,i}$ by a CES production function

$$\max_{Y_t, Y_{t,i}} \left\{ P_t\, Y_t - \sum_{i=1}^{N} P_{t,i}\, Y_{t,i} \right\}$$

subject to

$$Y_t = \left(\sum_{i=1}^{N} \left(Y_{t,i}\right)^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}}$$

Aggregator II

- Dividing the second FOC by the first gives

$$\frac{P_{t,i}}{P_t} = \left(\frac{Y_t}{Y_{t,i}}\right)^{\frac{1}{\theta}} \tag{26}$$

- This can be rearranged to get the demand function for the product of firm i

$$Y_{t,i} = d\left(P_{t,i}\right) = \left(\frac{P_{t,i}}{P_t}\right)^{-\theta} Y_t \tag{27}$$

Distributors' behavior

- Distribute the intermediate good $Y_{t,i}$
- They maximize profits using the demand function

$$Y_{t,i} = d\left(P_{t,i}\right) = \left(\frac{P_{t,i}}{P_t}\right)^{-\theta} Y_t \tag{28}$$

- They may charge a markup, so their input price $P^Y_{t,i}$ may be different from their selling price $P_{t,i}$
- They take Y_t and P_t as given

Adjustment cost

- Distributors face price adjustment cost
- Its exact form is

$$\Xi_t = \frac{\xi}{2}\,\bar{P}_{t,i}\,\bar{Y}_{t,i}\left(\log\left(\frac{P_{t,i}}{P_{t-1,i}}\right) - \log\left(\frac{P_{t-1}}{P_{t-2}}\right)\right)^2 \qquad (29)$$

- The cost is not zero if the distributor wants to increase prices more/less than previous *aggregate* inflation
- The bar over a variable means that its effect is ignored in the optimization

Markup and profit

- Adjustment costs and the market power will lead to a markup over the input price
- We need to differentiate between producer and final price
- Producer price of product i will be $P^Y_{t,i}$
- The profit of the distributor of product i in period t will therefore be

$$\left(P_{t,i} - P^Y_{t,i}\right) d\left(P_{t,i}\right) - \Xi_t \tag{30}$$

The objective of distributors

- The profit function of the distributors is

$$\sum_{t=0}^{\infty} \beta^t \Lambda_t \left(\left(P_{t,i} - P^Y_{t,i}\right) d\left(P_{t,i}\right) - \Xi_t \right) \tag{31}$$

- The only decision variable is $P_{t,i}$
- Again, we assume that optimizers own the distributors, so they use their owners' discount factor

The distributors' equation

- With the notation $ADJ_t = \xi \left(\log\left(\frac{P_{t,i}}{P_{t-1,i}}\right) - \log\left(\frac{P_{t-1}}{P_{t-2}}\right) \right)$
- Introduce the markup $\mu = \frac{\theta}{\theta - 1}$

$$\mu \, \frac{P^Y_{t,i}}{P_{t,i}} = 1 + (\mu - 1) \left(ADJ_t - \beta \, \frac{\Lambda_{t+1} \, P_{t+1,i} \, Y_{t+1,i}}{\Lambda_t \, P_{t,i} \, Y_{t,i}} \, ADJ_{t+1} \right) \qquad (32)$$

- And after the substitution $\Lambda_t = \bar{\Lambda}_t \, P_t$ this becomes

$$\mu \, \frac{P^Y_{t,i}}{P_{t,i}} = 1 + (\mu - 1) \left(ADJ_t - \beta \, \frac{\Lambda_{t+1} \, Y_{t+1,i}}{\Lambda_t \, Y_{t,i}} \, ADJ_{t+1} \right) \qquad (33)$$

Basic good producers

- Same as the producers before
- They produce from capital and labor, and sell for $P^Y_{t,i}$
- Therefore

$$RP^K_t \, K_{t-1,i} = \alpha \, \frac{P^Y_{t,i}}{P_{t,i}} \, Y_{t,i} \tag{34}$$

$$WR_t \, L_{t,i} = (1-\alpha) \, \frac{P^Y_{t,i}}{P_{t,i}} \, Y_{t,i} \tag{35}$$

$$Y_{t,i} = \left(K_{t-1,i}\right)^{\alpha} \, \left(A_t \, L_{t,i}\right)^{1-\alpha} \tag{36}$$

Equilibrium

- In equilibrium, all prices are equal, and therefore all quantities are equal as well (with suitable scaling in the CES aggregation function)
- At the end we get one new equation in the model

$$\mu \, RP_t^Y = 1 + (\mu - 1)\left(ADJ_t - \beta \, \frac{\Lambda_{t+1} \, Y_{t+1}}{\Lambda_t \, Y_t} \, ADJ_{t+1} \right) \tag{M.2}$$

- And the new price pressure variable will be $RP_t^Y = \frac{P_t^Y}{P_t}$

The Phillips curve I

- To see that this is indeed the New-Keynesian Phillips curve, let's again rearrange

$$\beta \frac{\Lambda_{t+1} Y_{t+1}}{\Lambda_t Y_t} ADJ_{t+1} - ADJ_t = \frac{\mu RP_t^Y - 1}{\mu - 1} \tag{37}$$

- Note that $\frac{\Lambda_{t+1} Y_{t+1}}{\Lambda_t Y_t} \approx 1$ around the steady state, therefore

$$\beta ADJ_{t+1} - ADJ_t = \frac{\mu RP_t^Y - 1}{\mu - 1} \tag{38}$$

in the log-linearized equation

The Phillips curve II

- Substitute the definition if ADJ_t

$$\beta\,\xi\left(\log\left(\mathrm{d}P_{t+1}\right)-\log\left(\mathrm{d}P_{t}\right)\right)-\xi\left(\log\left(\mathrm{d}P_{t}\right)-\log\left(\mathrm{d}P_{t-1}\right)\right)= \\ =\frac{\mu\,RP^{Y}_{t}-1}{\mu-1} \tag{39}$$

- Divide by ξ and collect terms on the left hand side

$$\beta\ \log\left(\mathrm{d}P_{t+1}\right)-(1+\beta)\log\left(\mathrm{d}P_{t}\right)+\log\left(\mathrm{d}P_{t-1}\right)= \\ =\frac{\mu\,RP^{Y}_{t}-1}{\xi\left(\mu-1\right)} \tag{40}$$

The Phillips curve III

- Rearrange to current inflation on the left hand side, and divide by $1 + \beta$

$$\log(\mathrm{d}P_t) = \frac{\beta}{1+\beta} \log(\mathrm{d}P_{t+1}) + \frac{1}{1+\beta} \log(\mathrm{d}P_{t-1}) + \frac{1}{(1+\beta)\,\xi\,(\mu-1)} \left(\mu\, RP_t^Y - 1\right) \tag{41}$$

- Using log-approximation of the relative price we get

$$\log(\mathrm{d}P_t) = \frac{\beta}{1+\beta} \log(\mathrm{d}P_{t+1}) + \frac{1}{1+\beta} \log(\mathrm{d}P_{t-1}) + \frac{1}{(1+\beta)\,\xi\,(\mu-1)} \log\left(\mu\, RP_t^Y\right) \tag{42}$$

The Phillips curve IV

- And the aggregate form of labor demand gives

$$\log\left(\mu\, RP_t^Y\right) = \log\left(\mu\, \frac{WR_t}{(1-\alpha)\, \frac{Y_t}{L_t}}\right) \tag{43}$$

relating RP_t^Y to the marginal cost of production

The New-Keynesian model

Model equations

$$\Lambda_t = \beta \, \Lambda_{t+1} R_t \tag{1}$$

$$C_t^{-\sigma} = \Lambda_t \tag{2}$$

$$\theta \, L_t^{\eta} = \Lambda_t \, WR_t \tag{3}$$

$$\Lambda_t = \beta \, \Lambda_{t+1} \left(RP_{t+1}^{K} + (1 - \delta) \right) \tag{4}$$

$$K_t = (1 - \delta) \, K_{t-1} + I_t \tag{5}$$

$$RP_t^K \, K_{t-1} = \alpha \, RP_t^Y \, Y_t \tag{6}$$

$$WR_t \, L_t = (1 - \alpha) \, RP_t^Y \, Y_t \tag{7}$$

$$Y_t = K_{t-1}^{\alpha} \, (A_t \, L_t)^{1-\alpha} \tag{8}$$

$$\begin{aligned} \log(\mathrm{d}P_t) &= \frac{\beta}{1+\beta} \log(\mathrm{d}P_{t+1}) + \frac{1}{1+\beta} \log(\mathrm{d}P_{t-1}) \\ &\quad + \frac{1}{(1+\beta)\,\xi\,(\mu - 1)} \log\left(\mu \, RP_t^Y\right) \end{aligned} \tag{9}$$

$$Y_t = C_t + I_t \tag{10}$$

Simplified model equations

$$C_t^{-\sigma} = \beta\, C_{t+1}^{-\sigma} R_t \tag{1}$$

$$\theta\, L_t^{\eta} = C_t^{-\sigma}\, WR_t \tag{2}$$

$$R_t = RP_{t+1}^{K} + (1-\delta) \tag{3}$$

$$K_t = (1-\delta)\, K_{t-1} + I_t \tag{4}$$

$$RP_t^{K}\, K_{t-1} = \alpha\, RP_t^{Y}\, Y_t \tag{5}$$

$$WR_t\, L_t = (1-\alpha)\, RP_t^{Y}\, Y_t \tag{6}$$

$$Y_t = K_{t-1}^{\alpha}\, \left(A_t\, L_t\right)^{1-\alpha} \tag{7}$$

$$\begin{aligned}\log\left(\mathrm{d}P_t\right) = {} & \frac{\beta}{1+\beta}\, \log\left(\mathrm{d}P_{t+1}\right) + \frac{1}{1+\beta}\, \log\left(\mathrm{d}P_{t-1}\right) \\ & + \frac{1}{(1+\beta)\,\xi\,(\mu-1)}\, \log\left(\mu\, RP_t^{Y}\right)\end{aligned} \tag{8}$$

$$Y_t = C_t + I_t \tag{9}$$

Disclaimer